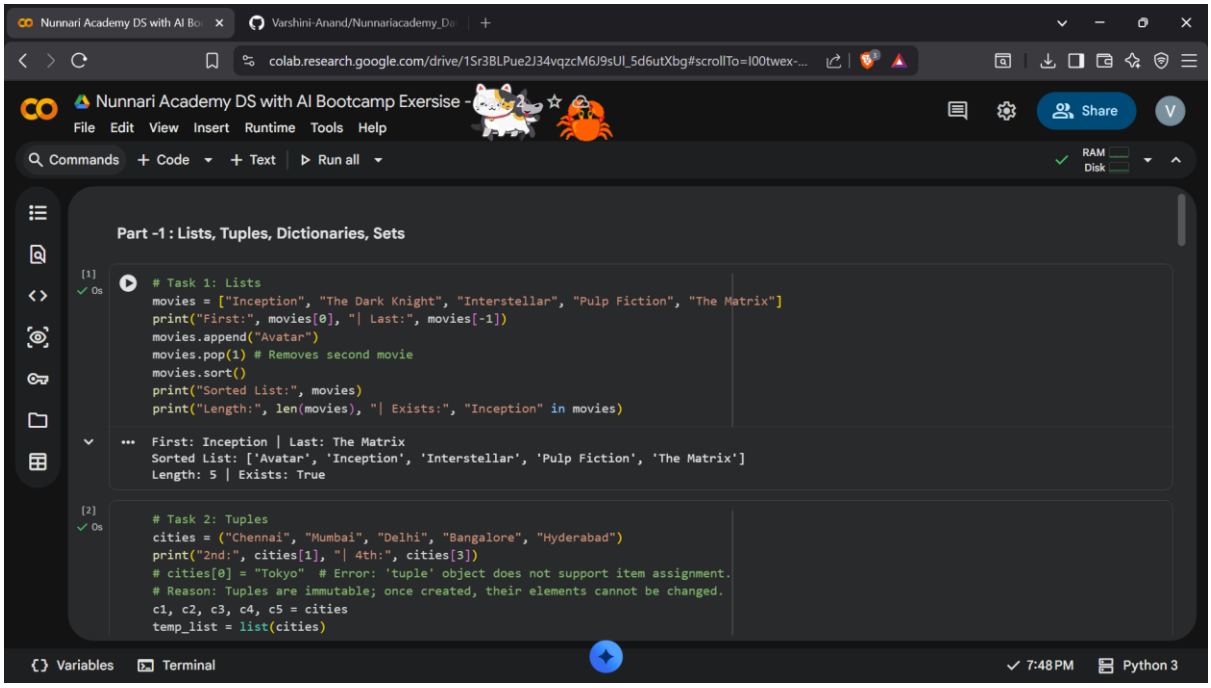
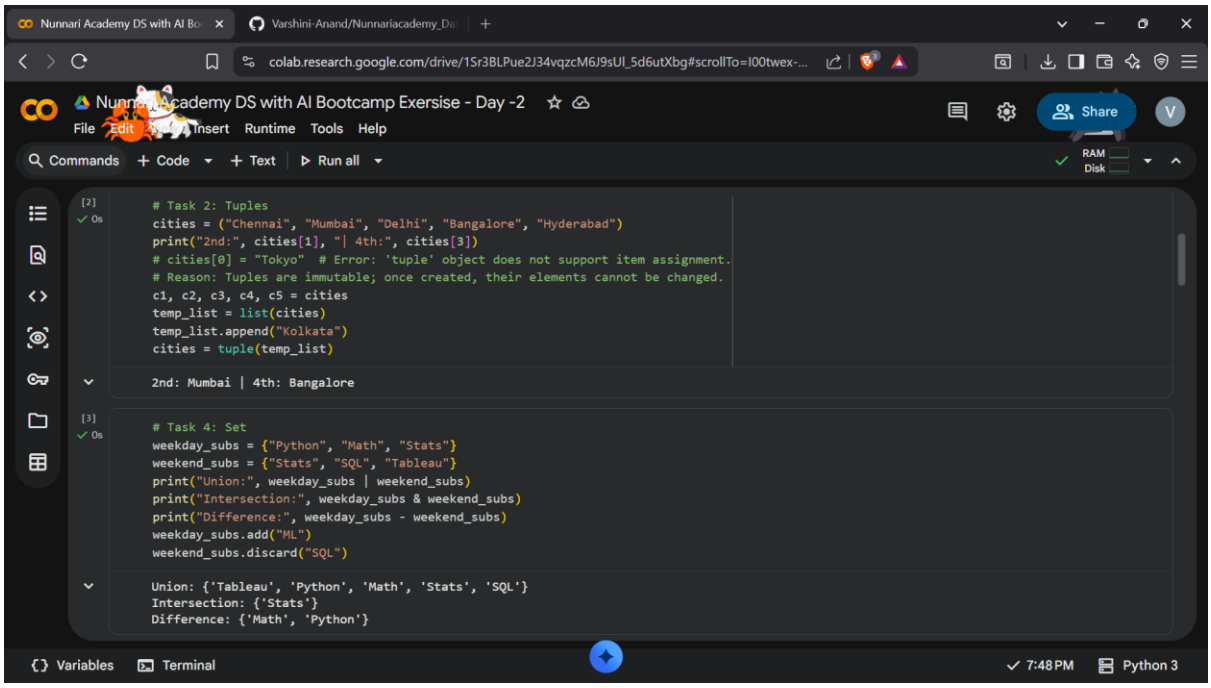
Nunnari Academy DS with AI Bootcamp Day – 2

Day – 2 work exercises:





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```
Difference: {'Math', 'Python'}
```

Part -2 : Functions, Logic and Loops

[4] 0s

```python
# Task 5 & 6: Functions & Default Parameters
def greet(name):
    print(f"Hello, {name}! Welcome to Day 2.")

def add_numbers(a, b):
    return a + b

def find_max(a, b, c):
    return max(a, b, c)

def is_even(n):
    return n % 2 == 0

def introduce(name, language="Python"):
    print(f"{name} is learning {language}.")

def calculate_discount(price, discount=10):
    return price * (1 - discount/100)
```

Variables Terminal 7:48 PM Python 3

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[5] 3s

```python
# Task 7: if / else / elif
num = int(input("Enter a number: "))
if num > 0: print("Positive")
elif num < 0: print("Negative")
else: print("Zero")
```

```
Enter a number: 45
Positive
```

[6] 0s

```python
# Task 8: For Loops
# Sum of 1 to 100
print("Sum 1-100:", sum(range(1, 101)))

# Fruits with index
fruits = ["Apple", "Banana", "Cherry"]
for i, fruit in enumerate(fruits):
    print(f"Index {i}: {fruit}")

# Count vowels
text = "Nunnari Academy"
vowels = sum(1 for char in text.lower() if char in "aeiou")
print(f"Vowel count: {vowels}")
```

```
Sum 1-100: 5050
Index 0: Apple
Index 1: Banana
Index 2: Cherry
Vowel count: 6
```

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Part - 3 : Mumpy Exercises

[7]

```python
import numpy as np

# Task 1: 1D Arrays
arr_1d = np.array([10, 20, 30, 40, 50])
zeros = np.zeros(10)
ones = np.ones(10)
range_arr = np.arange(1, 21)
lin_arr = np.linspace(0, 1, 7)
rand_ints = np.random.randint(1, 100, 8)
```

[8]

```python
# Task 2: 2D Arrays
zeros_2d = np.zeros((3, 3))
identity = np.eye(3)
full_sevens = np.full((4, 4), 7)
rand_floats = np.random.rand(3, 4)
nested_list_arr = np.array([[1,2,3], [4,5,6], [7,8,9]])
```

[9]

```python
# Task 3: Attributes
def print_attrs(arr):
    print(f"Dim: {arr.ndim}, Shape: {arr.shape}, Size: {arr.size}, Dtype: {arr.dtype}")

test_arr = np.random.rand(4, 5)
print_attrs(test_arr)
```

```
Dim: 2, Shape: (4, 5), Size: 20, Dtype: float64
```

[10]

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[10]

```python
# Task 4: Data Types
arr_int = np.array([1, 2, 3])
print("Before:", arr_int.dtype)
arr_float = arr_int.astype('float64')
print("After:", arr_float.dtype)
```

```
Before: int64
After: float64
```

[11]

```python
# Task 5: Special Functions
mid_arr = np.zeros(10)
mid_arr[len(mid_arr)//2] = 99
step_arr = np.arange(0, 51, 5)
v_stack = np.vstack(([1,2], [3,4]))
h_stack = np.hstack(([1,2], [3,4]))
```

[12]

```python
# Task 6: Indexing & Slicing
arr_15 = np.arange(1, 16)
print("First 5:", arr_15[:5])
print("Last 3:", arr_15[-3:])
print("Alternate:", arr_15[::2])

arr_3x3 = np.array([[1,2,3], [4,5,6], [7,8,9]])
print("Row 2, Col 3:", arr_3x3[1, 2])
print("Entire Col 3:", arr_3x3[:, 2])
```

```
First 5: [1 2 3 4 5]
Last 3: [13 14 15]
Alternate: [ 1  3  5  7  9 11 13 15]
Row 2, Col 3: 6
Entire Col 3: [3 6 9]
```

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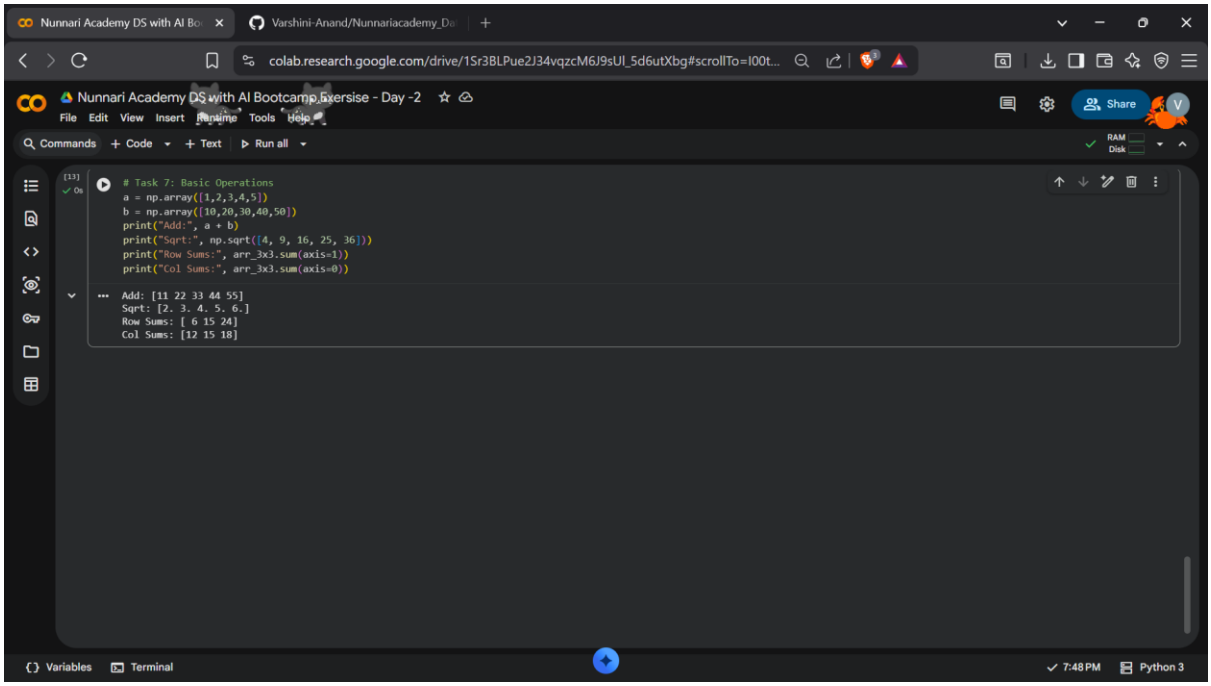
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# Task 7: Basic Operations
a = np.array([1,2,3,4,5])
b = np.array([10,20,30,40,50])
print("Add:", a + b)
print("Sqrt:", np.sqrt([4, 9, 16, 25, 36]))
print("Row Sums:", arr_3x3.sum(axis=1))
print("Col Sums:", arr_3x3.sum(axis=0))
Add: [11 22 33 44 55]
Sqrt: [2. 3. 4. 5. 6.]
Row Sums: [6 15 24]
Col Sums: [12 15 18]
Variables
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Python 3

Thank you..